

Assignment 1

Part 1

Part (e):

The smallest model came from int8 quantization while all three quantized models tied for highest accuracy.

I attempted to reduce the model size further by combining two techniques: knowledge distillation and full integer quantization. I trained a smaller student model using soft labels from the original teacher model, then converted the student model to an int8 TFLite model. This reduced the file size beyond the larger baseline and compressed models to 2.96 KB.

However, although the model size decreased, the classification accuracy dropped significantly to 0.8148. This means the further reduction was not successful under the requirement of preserving classification performance. The smaller student architecture likely did not have enough capacity to learn the class boundaries in the wine dataset, even with distillation. Additionally, int8 quantization reduces the numerical precision of weights and activations, which can further hurt performance when the model is already very small.

Therefore, further model size reduction is possible, but not without a major loss in classification performance. For this dataset and architecture, the model appears to have reached a practical compression limit.

Part 2:

1. [Dis] (5 pts) What types of “Learning Blocks” are available under the free-tier Edge Impulse account (e.g., supervised learning, classification and regression, unsupervised learning, clustering, self-supervised learning, etc.)? For each type, provide at least one example model that can be implemented using the free tier.

Learning blocks include supervised, unsupervised (clustering), and self-supervised models. Types available under the free-tier include:

- **Classification** (e.g. Neural-Network Classifier)
- **Regression** (e.g. Neural-Network Regressor)
- **Anomaly Detection** (e.g. K-Means Anomaly Detection, Gaussian Mixture Model (GMM))
- **Custom** (e.g. any user-defined model)

2. [Dis] (5 pts) List all the types of layers that can be used under the “Classifier” section in the free-tier Edge Impulse. Briefly explain the purpose of each layer identified.

There are 7 layers that can be used:

- **Input:** Receives the raw data and passes it forward; no computation is performed
- **Dense:** Used for general feature processing and as the final classification/regression output.
- **1D Convolution / pooling:** Learn features that take spatial information into account along a single dimension. Use this for raw data, or for DSP blocks that output spatial data
- **2D Convolution / pooling:** Learn features that take spatial information into account along two dimensions. Use this for raw data, or for DSP blocks that output spatial data.
- **Reshape:** Changes the shape of the data without altering its contents, often needed to align dimensions between different layer types.
- **Flatten:** Converts multi-dimensional data into a one-dimensional vector
- **Dropout:** Regularization technique that randomly sets a fraction of input units to zero during training, reducing overfitting.

3. [Dis] (5 pts) What types of model compression options are available under the “Deployment” section in the free-tier Edge Impulse? List all available methods.

Compression options include:

- **TFLite vs. EON Compiler:** Selects the inference engine. TFLite uses TensorFlow Lite for Microcontrollers; EON Compiler generates highly optimised C++ code that can further reduce RAM and flash usage.
- **Quantized (int8) vs. Unoptimized (float32):** Converts the model’s weights and activations from 32-bit floating-point to 8-bit integers

4. [Dis] (5 pts) What input modalities are supported for synthetic data generation in the free-tier version of Edge Impulse? For each modality, list the synthetic data generation methods/algorithms provided. Briefly explain why synthetic data is important in machine learning applications.

Supported input modalities include:

- Image
 - DALL-E Image Generation
 - Black Forest Labs FLUX-pro
- Audio
 - Whisper Synthetic Voice Generator

Synthetic data addresses data scarcity by generating large, diverse datasets for scenarios where real-world data is expensive, time-consuming, or physically impossible to obtain. Because synthetic data contains no real personal information, it also offers a privacy-safe alternative. Synthetic data accelerates rapid prototyping and experimentation, as teams can produce unlimited, perfectly labeled examples on demand without waiting for manual collection. It further improves model robustness by injecting controlled variations that act as a powerful form of augmentation, helping models generalize better to real-world conditions. Finally, synthetic generation can rebalance skewed datasets by creating equal numbers of samples for every class, mitigating class imbalance and leading to fairer, more accurate models